

Statistic and ProbabilityChapter 2 • ExercisesExercise 2-3

Total pendrive = 247

Red = 82

Blue = 75

Yellow = 90

$$a) P(\text{Red}) = \frac{n(\text{Red})}{N} = \frac{82}{247}$$

$$d) P(A | \text{blue}) = \frac{30}{75}$$

$$b) P(\text{blue} | B) = \frac{25}{75}$$

$$c) P(\text{blue}) = \frac{n(\text{blue})}{N} = \frac{75}{247}$$

$$e) P(\text{yellow} | C) = \frac{30}{90}$$

$$f) P(B \cap C) = \frac{n(74+80)}{247} = \frac{154}{247}$$

Exercise 2-4

Q1: $P(\text{wired} \cap \text{silver}) = 600/2465 \neq 0$. so they are not mutually exclusive

Q2: Are the events "Wireless" and "Orange" independent?

1st option: $P(\text{wireless} | \text{orange}) = P(\text{wireless})$

2nd option: $P(\text{orange} | \text{wireless}) = P(\text{orange})$

$$P(\text{orange}) = 710/2465 = 142/493$$

We can see that, $42/237$ is not equal to $142/493$.

Therefore, we can say that BOTH events wireless and orange are not

INDEPENDENT = they both are dependent.

Exercise 2-5

	P=40	Q	TOTAL
Defective = 20	8	$20-8=12$	20
Non Defective	$40-8=32$	$60-12=48$	80
TOTAL	40	60	100

Exercise 2-6

a.

$$IT = 0.80 \quad \text{pass both} = P(I|S) = \frac{0.75}{0.78} = 0.9615$$

$$S = 0.78$$

$$b) P(S|IT) = \frac{0.75}{0.8} = 0.9375$$

Exercise 2-9

$$a) P(\text{all upgraded}) = P(U) \cdot P(U) \cdot P(U) = (0.25)^3 \\ = (0.25) \cdot (0.25) \cdot (0.25) = 0.015625$$

b) ~~error~~

$$\bar{U} = 1.00 - 0.75 = 0.25$$

$$P_{\bar{U}} = P(\bar{U}) \cdot P(U) \cdot P(U) \\ = (0.75) \cdot (0.25) \cdot (0.25) = 0.0469$$

$$c) P(\bar{U})^3 = (0.75)^3 = 0.421875 \\ = 3[0.75^2 \times 0.25] = 0.421875 \\ = 0.421875 + 0.421875 = 0.84375$$

Exercise 2.11

$$U = \frac{N+T-N}{100} = \frac{54+69-35}{100} = \frac{88}{100} = 0.88$$

Exercise 2.12

A = fail B = longer / slow

A 0.35 B 0.35

$A|B = A$ $B|A = B$

$\bar{A} = 0.65$ $\bar{B} = 0.65$